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Gate structure and method of forming same

Abstract

A semiconductor device and a method of forming the same are provided. A method includes forming a sacrificial gate over an active region of a substrate. The sacrificial gate is removed to form an opening. A gate dielectric layer is formed on sidewalls and a bottom of the opening. A first work function layer is formed over the gate dielectric layer in the opening. A first protective layer is formed over the first work function layer in the opening. A first etch process is performed to widen an upper portion of the opening. The opening is filled with a conductive material.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE (1) This application is a continuation of U.S. patent application Ser. No. 17/811,651, filed on Jul. 11, 2022, now U.S. Pat. No. 11,837,507, issued

Dec. 5, 2023, entitled “Gate Structure and Method of Forming Same,” which is a divisional of U.S. patent application Ser. No. 16/564,243, filed on Sep. 9, 2019, now U.S. Pat. No. 11,404,327 issued Aug. 2, 2022, entitled “Gate Structure and Method of Forming Same,” each application is hereby incorporated herein by reference.

BACKGROUND

(1) Semiconductor devices are used in a variety of electronic applications, such as, for example, personal computers, cell phones, digital cameras, and other electronic equipment. Semiconductor devices are typically fabricated by sequentially depositing insulating or dielectric layers, conductive layers, and semiconductor layers of material over a semiconductor substrate, and patterning the various material layers using lithography to form circuit components and elements thereon.

(2) The semiconductor industry continues to improve the integration density of various electronic components (e.g., transistors, diodes, resistors, capacitors, etc.) by continual reductions in minimum feature size, which allow more components to be integrated into a given area. However, as the minimum features sizes are reduced, additional problems arise that should be addressed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 illustrates an example of a FinFET in a three-dimensional view in accordance with some embodiments.

(3) FIGS. 2, 3, 4, 5, 6, 7, 8A, 8B, 9A, 9B, 10A, 10B, 10C, 10D, 11A, 11B, 12A, 12B, 13A, 13B, 14A, 14B, 15A, 15B, 15C, 15D, 16A, 16B, 16C, 16D, 17A, 17B, 17C, 17D, 18A, 18B, 19A, and 19B are top and cross-sectional views of intermediate stages in the manufacturing of a FinFET device in accordance with some embodiments.

(4) FIGS. 20P, 20N, 21P, 21N, 22P, 22N, 23P, 23N, 24P, 24N, 25P, 25N, 26P, 26N, 27P, and 27N are cross-sectional views of intermediate stages in the manufacturing of gate structures in accordance with some embodiments.

(5) FIGS. 28P, 28N, 29P, 29N, 30P, 30N, 31P, 31N, 32P, 32N, 33P, 33N, 34P, 34N, 35P, and 35N are cross-sectional views of intermediate stages in the manufacturing of gate structures in accordance with some embodiments.

(6) FIGS. 36P, 36N, 37P, 37N, 38P, 38N, 39P, 39N, 40P, 40N, 41P, and 41N are cross-sectional views of intermediate stages in the manufacturing of gate structures in accordance with some embodiments.

(7) FIG. 42 is a flow diagram illustrating a method of forming a gate structure in accordance with some embodiments.

(8) FIG. 43 is a flow diagram illustrating a method of forming a gate structure in accordance with some embodiments.

(9) FIG. 44 is a flow diagram illustrating a method of forming a gate structure in accordance with some embodiments.

DETAILED DESCRIPTION

(10) The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or

on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(11) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(12) Embodiments will be described with respect to a specific context, namely, a gate structure of a semiconductor device and a method of forming the same. Various embodiments presented herein are discussed in the context of a FinFET device formed using a gate-last process. Some embodiments contemplate aspects used in planar transistor devices, multiple-gate transistor devices, 2D transistor devices, gate-all-around transistor devices, nanowire transistor devices, or the like. Various embodiments discussed herein allow for avoiding gate structure damage, such as formation of seam void defects in a gate structure and oxidation of work function layers of the gate structure, during subsequent processes, such as a gate cut process. Accordingly, threshold voltage ($V_{sub,t}$) shift and gate resistance ($R_{sub,G}$) increase due to the oxidation of work function layers of the gate structure may be reduced or avoided. Furthermore, threshold voltage ($V_{sub,t}$) roll-off may be reduced or avoided. Various embodiments discussed herein further allow for gate dielectric defect reduction and avoiding gate dielectric leakage and time-dependent gate dielectric breakdown (TDDB).

(13) FIG. 1 illustrates an example of a FinFET in a three-dimensional view, in accordance with some embodiments. The FinFET comprises a fin 52 on a substrate 50 (e.g., a semiconductor substrate). Isolation regions 56 are disposed in the substrate 50, and the fin 52 protrudes above and from between neighboring isolation regions 56. Although the isolation regions 56 are described/illustrated as being separate from the substrate 50, as used herein the term “substrate” may be used to refer to just the semiconductor substrate or a semiconductor substrate inclusive of isolation regions. Additionally, although the fin 52 is illustrated as a single, continuous material as the substrate 50, the fin 52 and/or the substrate 50 may comprise a single material or a plurality of materials. In this context, the fin 52 refers to the portion extending between the neighboring isolation regions 56.

(14) A gate dielectric layer 92 is along sidewalls and over a top surface of the fin 52, and a gate electrode 94 is over the gate dielectric layer 92. Source/drain regions 82 are disposed in opposite sides of the fin 52 with respect to the gate dielectric layer 92 and the gate electrode 94. FIG. 1 further illustrates reference cross-sections that are used in later figures. Cross-section A-A is along a longitudinal axis of the gate electrode 94 and in a direction, for example, perpendicular to a direction of a current flow between the source/drain regions 82 of the FinFET. Cross-section B-B is perpendicular to cross-section A-A and is along a longitudinal axis of the fin 52 and in a direction of, for example, a current flow between the source/drain regions 82 of the FinFET. Cross-section C-C is parallel to cross-section A-A and extends through the source/drain region 82 of the FinFET. Subsequent figures refer to these reference cross-sections for clarity.

(15) FIGS. 2, 3, 4, 5, 6, 7, 8A, 8B, 9A, 9B, 10A, 10B, 10C, 10D, 11A, 11B, 12A, 12B, 13A, 13B, 14A, 14B, 15A, 15B, 15C, 15D, 16A, 16B, 16C, 16D, 17A, 17B, 17C, 17D, 18A, 18B, 19A, and 19B are top and cross-sectional views of intermediate stages in the manufacturing of a FinFET

device in accordance with some embodiments. FIGS. 2 through 7 illustrate cross-sectional views along the reference cross-section A-A illustrated in FIG. 1, except for multiple fins/FinFETs. FIGS. 8A-14A, 18A, and 19A are illustrated along the reference cross-section A-A illustrated in FIG. 1 and FIGS. 8B-14B, 18B, and 19B are illustrated along the reference cross-section B-B illustrated in FIG. 1, except for multiple fins/FinFETs. FIGS. 10C and 10D are illustrated along the reference cross-section C-C illustrated in FIG. 1, except for multiple fins/FinFETs. FIGS. 15D-17D illustrate top views. FIGS. 15A-17A are illustrated along the cross-section A-A illustrated in FIGS. 15D-17D, respectively. FIGS. 15B-17B are illustrated along the cross-section B-B illustrated in FIGS. 15D-17D, respectively. FIGS. 15C-17C are illustrated along the cross-section C-C illustrated in FIGS. 15D-17D, respectively.

(16) In FIG. 2, a substrate 50 is provided. The substrate 50 may be a semiconductor substrate, such as a bulk semiconductor, a semiconductor-on-insulator (SOI) substrate, or the like, which may be doped (e.g., with a p-type or an n-type dopant) or undoped. The substrate 50 may be a wafer, such as a silicon wafer. Generally, an SOI substrate is a layer of a semiconductor material formed on an insulator layer. The insulator layer may be, for example, a buried oxide (BOX) layer, a silicon oxide layer, or the like. The insulator layer is provided on a substrate, typically a silicon or glass substrate. Other substrates, such as a multi-layered or gradient substrate may also be used. In some embodiments, the semiconductor material of the substrate 50 may include silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof.

(17) The substrate 50 has a region 50N and a region 50P. The region 50N can be for forming n-type devices, such as NMOS transistors, e.g., n-type FinFETs. The region 50P can be for forming p-type devices, such as PMOS transistors, e.g., p-type FinFETs. The region 50N may be physically separated from the region 50P (as illustrated by a divider 51), and any number of device features (e.g., other active devices, doped regions, isolation structures, etc.) may be disposed between the region 50N and the region 50P.

(18) In FIG. 3, fins 52 are formed in the substrate 50. The fins 52 are semiconductor strips. In some embodiments, the fins 52 may be formed in the substrate 50 by etching trenches in the substrate 50. The etching may be any acceptable etch process, such as a reactive ion etch (RIE), neutral beam etch (NBE), the like, or a combination thereof. The etch process may be anisotropic.

(19) The fins may be formed by any suitable method. For example, the fins may be formed using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used as a mask to form the fins.

(20) In FIG. 4, an insulation material 54 is formed over the substrate 50 and between neighboring fins 52. The insulation material 54 may be an oxide, such as silicon oxide, a nitride, a combination thereof, or the like, and may be formed by a high density plasma chemical vapor deposition (HDP-CVD), a flowable CVD (FCVD) (e.g., a CVD-based material deposition in a remote plasma system and post curing to make it convert to another material, such as an oxide), a combination thereof, or the like. Other insulation materials formed by any acceptable process may be used. In the illustrated embodiment, the insulation material 54 is silicon oxide formed by a FCVD process. An anneal process may be performed once the insulation material is formed. In an embodiment, the insulation material 54 is formed such that excess insulation material 54 covers the fins 52.

Although the insulation material 54 is illustrated as a single layer, some embodiments may utilize

multiple layers. For example, in some embodiments, a liner (not shown) may first be formed along a surface of the substrate **50** and the fins **52**. Thereafter, a fill material, such as those discussed above may be formed over the liner.

(21) In FIG. **5**, a removal process is applied to the insulation material **54** to remove excess portions of the insulation material **54** over the fins **52**. In some embodiments, a planarization process, such as a chemical mechanical polish (CMP) process, an etch back process, combinations thereof, or the like, may be utilized. The planarization process exposes the fins **52** such that top surfaces of the fins **52** and the insulation material **54** are level after the planarization process is completed.

(22) In FIG. **6**, the insulation material **54** (see FIG. **5**) is recessed to form Shallow Trench Isolation (STI) regions **56**. The insulation material **54** is recessed such that upper portions of fins **52** in the regions **50N** and **50P** protrude from between neighboring STI regions **56**. Further, the top surfaces of the STI regions **56** may have a flat surface as illustrated, a convex surface, a concave surface (such as dishing), or a combination thereof. The top surfaces of the STI regions **56** may be formed flat, convex, and/or concave by an appropriate etch. The STI regions **56** may be recessed using an acceptable etching process, such as one that is selective to the material of the insulation material **54** (e.g., etches the material of the insulation material **54** at a faster rate than the material of the fins **52**). For example, a chemical oxide removal with a suitable etch process using, for example, dilute hydrofluoric (dHF) acid may be used.

(23) The process described with respect to FIGS. **2** through **6** is just one example of how the fins **52** may be formed. In some embodiments, the fins may be formed by an epitaxial growth process. For example, a dielectric layer can be formed over a top surface of the substrate **50**, and trenches can be etched through the dielectric layer to expose the underlying substrate **50**. Homoepitaxial structures can be epitaxially grown in the trenches, and the dielectric layer can be recessed such that the homoepitaxial structures protrude from the dielectric layer to form fins. Additionally, in some embodiments, heteroepitaxial structures can be used for the fins **52**. For example, the fins **52** in FIG. **5** can be recessed, and a material different from the fins **52** may be epitaxially grown over the recessed fins **52**. In such embodiments, the fins **52** comprise the recessed material as well as the epitaxially grown material disposed over the recessed material. In an even further embodiment, a dielectric layer can be formed over a top surface of the substrate **50**, and trenches can be etched through the dielectric layer. Heteroepitaxial structures can then be epitaxially grown in the trenches using a material different from the substrate **50**, and the dielectric layer can be recessed such that the heteroepitaxial structures protrude from the dielectric layer to form the fins **52**. In some embodiments where homoepitaxial or heteroepitaxial structures are epitaxially grown, the epitaxially grown materials may be in situ doped during growth, which may obviate prior and subsequent implantations, although in situ and implantation doping may be used together.

(24) Still further, it may be advantageous to epitaxially grow a material in the region **50N** different from a material in the region **50P**. In various embodiments, upper portions of the fins **52** may be formed from silicon germanium ($\text{Si.sub.xGe.sub.1-x}$, where x can be in the range of 0 to 1), silicon carbide, pure or substantially pure germanium, a III-V compound semiconductor, a II-VI compound semiconductor, or the like. For example, the available materials for forming III-V compound semiconductor include, but are not limited to, InAs, AlAs, GaAs, InP, GaN, InGaAs, InAlAs, GaSb, AlSb, AlP, GaP, and the like.

(25) Further in FIG. **6**, appropriate wells (not shown) may be formed in the fins **52** and/or the substrate **50**. In some embodiments, a P well may be formed in the region **50N**, and an N well may be formed in the region **50P**. In some embodiments, a P well or an N well are formed in both the region **50N** and the region **50P**. In the embodiments with different well types, the different implant steps for the region **50N** and the region **50P** may be achieved using a photoresist or other masks (not shown). For example, a photoresist may be formed over the fins **52** and the STI regions **56** in the region **50N**. The photoresist is patterned to expose the region **50P** of the substrate **50**. The photoresist can be formed by using a spin-on technique and can be patterned using acceptable

photolithography techniques. Once the photoresist is patterned, an n-type impurity implantation is performed in the region **50P**, and the photoresist may act as a mask to substantially prevent n-type impurities from being implanted into the region **50N**. The n-type impurities may be phosphorus, arsenic, antimony, or the like, implanted in the region to a dose of equal to or less than $10 \times 10^{15} \text{ cm}^{-2}$, such as between about $10 \times 10^{12} \text{ cm}^{-2}$ and about $10 \times 10^{15} \text{ cm}^{-2}$. In some embodiments, the n-type impurities may be implanted at an implantation energy of about 1 keV to about 10 keV. After the implantation, the photoresist is removed, such as by an acceptable ashing process followed by a wet clean process.

(26) Following the implantation of the region **50P**, a photoresist is formed over the fins **52** and the STI regions **56** in the region **50P**. The photoresist is patterned to expose the region **50N** of the substrate **50**. The photoresist can be formed by using a spin-on technique and can be patterned using acceptable photolithography techniques. Once the photoresist is patterned, a p-type impurity implantation may be performed in the region **50N**, and the photoresist may act as a mask to substantially prevent p-type impurities from being implanted into the region **50P**. The p-type impurities may be boron, BF₂, indium, or the like, implanted in the region to a dose of equal to or less than $10 \times 10^{15} \text{ cm}^{-2}$, such as between about $10 \times 10^{12} \text{ cm}^{-2}$ and about $10 \times 10^{15} \text{ cm}^{-2}$. In some embodiments, the p-type impurities may be implanted at an implantation energy of about 1 keV to about 10 keV. After the implantation, the photoresist may be removed, such as by an acceptable ashing process followed by a wet clean process.

(27) After performing the implantations of the region **50N** and the region **50P**, an anneal may be performed to activate the p-type and/or n-type impurities that were implanted. In some embodiments, the grown materials of epitaxial fins may be in situ doped during growth, which may obviate the implantations, although in situ and implantation doping may be used together.

(28) In FIG. 7, a dummy dielectric layer **60** is formed on the fins **52**. The dummy dielectric layer **60** may be, for example, silicon oxide, silicon nitride, a combination thereof, or the like, and may be deposited or thermally grown according to acceptable techniques. A dummy gate layer **62** is formed over the dummy dielectric layer **60**, and a mask layer **64** is formed over the dummy gate layer **62**. The dummy gate layer **62** may be deposited over the dummy dielectric layer **60** and then planarized using, for example, a CMP process. The mask layer **64** may be deposited over the dummy gate layer **62**. The dummy gate layer **62** may be a conductive material and may be selected from a group including amorphous silicon, polycrystalline-silicon (polysilicon), poly-crystalline silicon-germanium (poly-SiGe), metallic nitrides, metallic silicides, metallic oxides, and metals. The dummy gate layer **62** may be deposited by physical vapor deposition (PVD), CVD, sputter deposition, or other techniques known and used in the art for depositing conductive materials. The dummy gate layer **62** may be made of other materials that have a high etching selectivity than materials of the STI regions **56**. The mask layer **64** may include, for example, SiN, SiON, a combination thereof, or the like. In this example, a single dummy gate layer **62** and a single mask layer **64** are formed across the region **50N** and the region **50P**. It is noted that the dummy dielectric layer **60** is shown covering only the fins **52** for illustrative purposes only. In some embodiments, the dummy dielectric layer **60** may be deposited such that the dummy dielectric layer **60** covers the STI regions **56**, extending between the dummy gate layer **62** and the STI regions **56**.

(29) FIGS. 8A, 8B, 9A, 9B, 10A, 10B, 10C, 10D, 11A, 11B, 12A, 12B, 13A, 13B, 14A, 14B, 15A, 15B, 15C, 15D, 16A, 16B, 16C, 16D, 17A, 17B, 17C, 17D, 18A, 18B, 19A, and 19B illustrate various additional steps in the manufacturing of a FinFET device. FIGS. 8A, 8B, 9A, 9B, 10A, 10B, 10C, 10D, 11A, 11B, 12A, 12B, 13A, 13B, 14A, 14B, 15A, 15B, 15C, 15D, 16A, 16B, 16C, 16D, 17A, 17B, 17C, 17D, 18A, 18B, 19A, and 19B illustrate features in either of the region **50N** and the region **50P**. For example, the structures illustrated in FIGS. 8A, 8B, 9A, 9B, 10A, 10B, 10C, 10D, 11A, 11B, 12A, 12B, 13A, 13B, 14A, 14B, 15A, 15B, 15C, 15D, 16A, 16B, 16C, 16D, 17A, 17B, 17C, 17D, 18A, 18B, 19A, and 19B may be applicable to both the region **50N** and the region **50P**. Differences (if any) in the structures of the region **50N** and the region **50P** are described

in the text accompanying each figure.

(30) In FIGS. **8A** and **8B**, the mask layer **64** (see FIG. **7**) may be patterned using acceptable photolithography and etching techniques to form masks **74**. The pattern of the masks **74** then may be transferred to the dummy gate layer **62** to form dummy gates **72**. In some embodiments (not illustrated), the pattern of the masks **74** may also be transferred to the dummy dielectric layer **60** by an acceptable etching technique. The dummy gates **72** cover respective channel regions **58** of the fins **52**. The pattern of the masks **74** may be used to physically separate each of the dummy gates **72** from adjacent dummy gates. The dummy gates **72** may also have a lengthwise direction substantially perpendicular to the lengthwise direction of respective fins **52**. As described below in greater detail, the dummy gates **72** are sacrificial gates and are subsequently replaced by replacement gates. Accordingly, dummy gates **72** may also be referred to as sacrificial gates. In other embodiments, some of the dummy gates **72** are not replaced and remain in the final structure of the resulting FinFET device.

(31) Further in FIGS. **8A** and **8B**, gate seal spacers **80** may be formed on exposed surfaces of the dummy gates **72**, the masks **74**, and/or the fins **52**. A thermal oxidation or a deposition followed by an anisotropic etch may form the gate seal spacers **80**. The gate seal spacers **80** may comprise silicon oxide, silicon nitride, SiCN, SiOC, SiOCN, a combination thereof, or the like. After the formation of the gate seal spacers **80**, implants for lightly doped source/drain (LDD) regions (not explicitly illustrated) may be performed. In the embodiments with different device types, similar to the implants discussed above in FIG. **6**, a mask, such as a photoresist, may be formed over the region **50N**, while exposing the region **50P**, and appropriate type (e.g., p-type) impurities may be implanted into the exposed fins **52** in the region **50P**. The mask may then be removed.

Subsequently, a mask, such as a photoresist, may be formed over the region **50P**, while exposing the region **50N**, and appropriate type impurities (e.g., n-type) may be implanted into the exposed fins **52** in the region **50N**. The mask may then be removed. The n-type impurities may be the any of the n-type impurities previously discussed, and the p-type impurities may be the any of the p-type impurities previously discussed. The lightly doped source/drain regions may have a dose of impurities of from about 10^{12} cm⁻² to about 10^{16} cm⁻². In some embodiments, the suitable impurities may be implanted at an implantation energy of about 1 keV to about 10 keV. An anneal may be used to activate the implanted impurities.

(32) In FIGS. **9A** and **9B**, gate spacers **86** are formed on the gate seal spacers **80** along sidewalls of the dummy gates **72** and the masks **74**. The gate spacers **86** may be formed by conformally depositing an insulating material and subsequently anisotropically etching the insulating material. The insulating material of the gate spacers **86** may comprise silicon oxide, silicon nitride, SiCN, SiOC, SiOCN, a combination thereof, or the like. In some embodiments, the gate spacers **86** may comprise a plurality of layers (not shown), such that the layers comprise different materials.

(33) In FIGS. **10A** and **10B**, epitaxial source/drain regions **82** are formed in the fins **52** to exert stress in the respective channel regions **58**, thereby improving device performance. The epitaxial source/drain regions **82** are formed in the fins **52** such that each dummy gate **72** is disposed between respective neighboring pairs of the epitaxial source/drain regions **82**. In some embodiments, the epitaxial source/drain regions **82** may extend into, and may also penetrate through, the fins **52**. In some embodiments, the gate spacers **86** are used to separate the epitaxial source/drain regions **82** from the dummy gates **72** by an appropriate lateral distance so that the epitaxial source/drain regions **82** do not short out subsequently formed gates of the resulting FinFET device.

(34) The epitaxial source/drain regions **82** in the region **50N** may be formed by masking the region **50P** and etching source/drain regions of the fins **52** in the region **50N** to form recesses in the fins **52**. Then, the epitaxial source/drain regions **82** in the region **50N** are epitaxially grown in the recesses. The epitaxial source/drain regions **82** may include any acceptable material, such as appropriate for n-type FinFETs. For example, if the fin **52** is silicon, the epitaxial source/drain

regions **82** in the region **50N** may include materials exerting a tensile strain in the channel region **58**, such as silicon, SiC, SiCP, SiP, a combination thereof, or the like. The epitaxial source/drain regions **82** in the region **50N** may have surfaces raised from respective surfaces of the fins **52** and may have facets.

(35) The epitaxial source/drain regions **82** in the region **50P** may be formed by masking the region **50N** and etching source/drain regions of the fins **52** in the region **50P** to form recesses in the fins **52**. Then, the epitaxial source/drain regions **82** in the region **50P** are epitaxially grown in the recesses. The epitaxial source/drain regions **82** may include any acceptable material, such as appropriate for p-type FinFETs. For example, if the fin **52** is silicon, the epitaxial source/drain regions **82** in the region **50P** may comprise materials exerting a compressive strain in the channel region **58**, such as SiGe, SiGeB, Ge, GeSn, a combination thereof, or the like. The epitaxial source/drain regions **82** in the region **50P** may also have surfaces raised from respective surfaces of the fins **52** and may have facets.

(36) The epitaxial source/drain regions **82** and/or the fins **52** may be implanted with dopants to form source/drain regions, similar to the process previously discussed for forming lightly-doped source/drain regions, followed by an anneal. The source/drain regions may have an impurity concentration of between about 10^{19} cm⁻³ and about 10^{21} cm⁻³. The n-type and/or p-type impurities for source/drain regions may be any of the impurities previously discussed. In some embodiments, the epitaxial source/drain regions **82** may be in situ doped during growth.

(37) As a result of the epitaxy processes used to form the epitaxial source/drain regions **82** in the region **50N** and the region **50P**, upper surfaces of the epitaxial source/drain regions have facets which expand laterally outward beyond sidewalls of the fins **52**. In some embodiments, these facets cause adjacent epitaxial source/drain regions **82** of a same FinFET to merge as illustrated by FIG. **10C**. In other embodiments, adjacent epitaxial source/drain regions **82** remain separated after the epitaxy process is completed as illustrated by FIG. **10D**.

(38) In FIGS. **11A** and **11B**, a first ILD **88** is deposited over the structure illustrated in FIGS. **10A** and **10B**. The first ILD **88** may be formed of a dielectric material, and may be deposited by any suitable method, such as CVD, plasma-enhanced CVD (PECVD), FCVD, a combination thereof, or the like. Dielectric materials may include Phospho-Silicate Glass (PSG), Boro-Silicate Glass (BSG), Boron-Doped Phospho-Silicate Glass (BPSG), undoped Silicate Glass (USG), or the like. Other insulation materials formed by any acceptable process may be also used. In some embodiments, a contact etch stop layer (CESL) **87** is disposed between the first ILD **88** and the epitaxial source/drain regions **82**, the masks **74**, and the gate spacers **86**. The CESL **87** may comprise a dielectric material, such as, silicon nitride, silicon oxide, silicon oxynitride, a combination thereof, or the like, having a different etch rate than the material of the overlying first ILD **88**.

(39) In FIGS. **12A** and **12B**, a planarization process, such as a CMP process, may be performed to level the top surface of the first ILD **88** with the top surfaces of the dummy gates **72** or the masks **74** (see FIGS. **11A** and **11B**). The planarization process may also remove the masks **74** on the dummy gates **72**, and portions of the gate seal spacers **80** and the gate spacers **86** along sidewalls of the masks **74**. After the planarization process, top surfaces of the dummy gates **72**, the gate seal spacers **80**, the gate spacers **86**, and the first ILD **88** are level with each other. Accordingly, the top surfaces of the dummy gates **72** are exposed through the first ILD **88**. In some embodiments, the masks **74** may remain, in which case the planarization process levels the top surface of the first ILD **88** with the top surfaces of the masks **74** (see FIGS. **11A** and **11B**).

(40) In FIGS. **13A** and **13B**, the dummy gates **72**, and the masks **74**, if present, are removed in an etching step(s), so that openings **90** are formed. Portions of the dummy dielectric layer **60** in the openings **90** may also be removed. In some embodiments, only the dummy gates **72** are removed and the dummy dielectric layer **60** remains and is exposed by the openings **90**. In some

embodiments, the dummy dielectric layer **60** is removed from the openings **90** in a first region of a die (e.g., a core logic region) and remains in openings **90** in a second region of the die (e.g., an input/output region). In some embodiments, the dummy gates **72** are removed by an anisotropic dry etch process. For example, the etching process may include a dry etch process using reaction gas(es) that selectively etch the dummy gates **72** without etching the first ILD **88** or the gate spacers **86**. Each opening **90** exposes a channel region **58** of a respective fin **52**. Each channel region **58** is disposed between neighboring pairs of the epitaxial source/drain regions **82**. During the removal, the dummy dielectric layer **60** may be used as an etch stop layer when the dummy gates **72** are etched. The dummy dielectric layer **60** may then be optionally removed after the removal of the dummy gates **72**.

(41) In FIGS. **14A** and **14B**, gate dielectric layers **92** and gate electrodes **94** are formed for replacement gates. The gate electrodes **94** and the gate dielectric layers **92** may be collectively referred to as a “gate stack.” The gate stacks may extend along sidewalls and top surfaces of the channel regions **58** of the fins **52**. In some embodiments, the gate stacks may be formed using a method described below with reference to FIGS. **20P**, **20N**, **21P**, **21N**, **22P**, **22N**, **23P**, **23N**, **24P**, **24N**, **25P**, **25N**, **26P**, **26N**, **27P**, and **27N**, and the detailed description of the method is provided at that time. In other embodiments, the gate stacks may be formed using a method described below with reference to FIGS. **28P**, **28N**, **29P**, **29N**, **30P**, **30N**, **31P**, **31N**, **32P**, **32N**, **33P**, **33N**, **34P**, **34N**, **35P**, and **35N**, and the detailed description of the method is provided at that time. In yet other embodiments, the gate stacks may be formed using a method described below with reference to FIGS. **36P**, **36N**, **37P**, **37N**, **38P**, **38N**, **39P**, **39N**, **40P**, **40N**, **41P**, and **41N**, and the detailed description of the method is provided at that time.

(42) FIGS. **15A**, **15B**, **15C**, **15D**, **16A**, **16B**, **16C**, **16D**, **17A**, **17B**, **17C**, and **17D**, illustrate a gate cut process in accordance with some embodiments. FIGS. **15D-17D** illustrate top views. FIGS. **15A-17A** are illustrated along the cross-section A-A illustrated in FIGS. **15D-17D**, respectively. FIGS. **15B-17B** are illustrated along the cross-section B-B illustrated in FIGS. **15D-17D**, respectively. FIGS. **15C-17C** are illustrated along the cross-section C-C illustrated in FIGS. **15D-17D**, respectively. In FIGS. **15A**, **15B**, **15C** and **15D**, a hard mask **128** is formed over the gate stack **92/94**. In some embodiments, the hard mask **128** may comprise one or more layers of a dielectric material, such as silicon oxide, titanium nitride, silicon nitride, silicon oxynitride, a combination thereof, or the like, and may be formed using CVD, ALD, a combination thereof, or the like. In an embodiment, the hard mask **128** includes a first mask layer **128A** comprising titanium nitride and a second mask layer **128B** comprising silicon nitride, with the second mask layer **128B** being formed over the first mask layer **128A**. The hard mask **128** is patterned to form openings **130** through the hard mask **128**. In some embodiments, the hard mask **128** may be patterned using suitable photolithography and etching processes. The etching processes may be wet or dry etching processes. The etching processes may be anisotropic etching processes. In some embodiments, the openings **130** may have shapes of elongated trenches, such that a lengthwise direction of each of the trenches is substantially parallel to a lengthwise direction of the fins **52** and is substantially perpendicular to a lengthwise direction of the gate stacks **92/94**.

(43) In FIGS. **16A**, **16B**, **16C** and **16D**, the hard mask **128** is used to pattern the gate stack **92/94**. In some embodiments, the gate stack **92/94** is patterned by one or more suitable etching processes using the hard mask **128** as an etch mask. The etching processes may include wet etching processes, dry etching processes, combinations thereof, or the like. The etching processes may be anisotropic etching processes. In some embodiments, the etching processes include only dry etching processes. In other embodiments, the etching processes include a combination of both dry etching and wet etching processes. In some embodiments when the etching processes include only dry etching processes, the etching processes are followed by wet cleaning process to clean off residues of the etching processes. In some embodiments, the dry etching processes may be performed using a mixture of etchants comprising Cl.sub.2, SiCl.sub.4, CH.sub.4, O.sub.2,

CF.sub.4, BCl.sub.3, Ar, a combination thereof, or the like. In some embodiments, the wet etching processes may be performed using a mixture of etchants comprising HCl, H.sub.2O.sub.2, (de-ionized) DI water, a combination thereof, or the like. In some embodiments, the wet cleaning process is performed using DI water, a mixture of DI water and dilute HCl, or the like. In some embodiments, the gate electrode **94** of the gate stacks **92/94** is removed by a dry etching process using a mixture of etchants comprising Cl.sub.2, SiCl.sub.4, O.sub.2, BCl.sub.3, a combination thereof, or the like. In some embodiments, the gate dielectric layer **92** of the gate stacks **92/94** is removed by a dry etching process using a mixture of etchants comprising BCl.sub.3, or the like. The patterning process forms openings **132**, such that the openings **132** extend through the gate stack **92/94** and into the isolation regions **56**. In some embodiments, the openings **132** have sloped sidewalls. In some embodiments, top portions of the openings **132** have a first width W.sub.1 and bottom portions of the openings **132** have a second width W.sub.2. In some embodiments, the first width W.sub.1 is greater than the second width W.sub.2. In some embodiments, the first width W.sub.1 is between about 14 nm and about 50 nm. In some embodiments, the second width W.sub.2 is between about 8 nm and about 30 nm. In some embodiments, a ratio W.sub.1/W.sub.2 is between about 1.1 and about 1.4. In some embodiments, the openings **132** extend below topmost surfaces the isolation regions **56** to a depth D.sub.1. In some embodiments, the depth D.sub.1 is between about 20 nm and about 55 nm.

(44) Further in FIGS. **16A**, **16B**, **16C** and **16D**, in some embodiments, patterned gate stacks **92/94** may be damaged by etchants used for forming the openings **132**. In some embodiments, the etchants may penetrate into the patterned gate stacks **92/94** from sidewalls of the openings **132** and may damage the patterned gate stacks **92/94**. By forming the gate stacks **92/94** as described below with reference to FIGS. **20P-41P** and **20N-41N**, seam void defects are not formed in the gate stacks **92/94** and the etchants of the gate cut process do not penetrate into the seam void defects to damage the gate stacks **92/94**.

(45) In FIGS. **17A**, **17B**, **17C**, and **17D**, a dielectric material **134** is formed in the openings **130** (see FIGS. **15A**, **15B**, **15C**, and **15D**) and the openings **132** (see FIGS. **16A**, **16B**, **16C**, and **16D**). In some embodiments, the dielectric material **134** silicon oxide, titanium nitride, silicon nitride, silicon oxynitride, a combination thereof, or the like, and may be formed using CVD, ALD, a combination thereof, or the like. In some embodiments, the dielectric material **134** overfills the openings **130** and **132** such that a portion of the dielectric material **134** is disposed over a topmost surface of the hard mask **128**.

(46) In FIGS. **18A** and **18B**, a planarization process is performed to remove the hard mask **128** (see FIGS. **17A**, **17B**, **17C**, and **17D**) and portions of the dielectric material **134** (see FIGS. **17A**, **17B**, **17C**, and **17D**) disposed above the topmost surface of the gate stack. In some embodiments, the planarization process may comprise a CMP process, an etch process, a combination thereof, or the like. Remaining portions of the dielectric material **134** form isolation regions **136** separating and electrically isolating adjacent portions of the patterned gate stack **92/94**. In the embodiment illustrated in FIGS. **18A** and **18B**, a portion of the patterned gate stacks **92/94** overlaps with two fins of the fins **52**. In other embodiments, portions of the patterned gate stacks **92/94** may overlap with one fin or more than two fins of the fins **52**.

(47) Further in FIGS. **18A** and **18B**, after performing the planarization process, a second ILD **108** is deposited over the first ILD **88** and the gate stacks **92/94** (including gate dielectric layers **92** and corresponding overlying gate electrodes **94**). In some embodiment, the second ILD **108** is a flowable film formed by a flowable CVD method. In some embodiments, the second ILD **108** is formed of a dielectric material such as PSG, BSG, BPSG, USG, a combination thereof, or the like, and may be deposited by any suitable method, such as CVD, PECVD, a combination thereof, or the like. In some embodiments, before the formation of the second ILD **108**, the gate stacks **92/94** are recessed, so that recesses are formed directly over the gate stacks **92/94** and between opposing portions of gate spacers **86**. Gate masks **96** comprising one or more layers of dielectric material,

such as silicon nitride, silicon oxynitride, a combination thereof, or the like, are filled in the recesses, followed by a planarization process to remove excess portions of the dielectric material extending over the first ILD **88**. The subsequently formed gate contacts **110** (see FIGS. **19A** and **19B**) penetrate through the respective gate mask **96** to contact the top surface of the respective recessed gate electrode **94**.

(48) In FIGS. **19A** and **19B**, gate contacts **110** and source/drain contacts **112** are formed through the second ILD **108** and the first ILD **88** in accordance with some embodiments. Openings for the source/drain contacts **112** are formed through the first ILD **88** and the second ILD **108**, and openings for the gate contacts **110** are formed through the second ILD **108** and the gate masks **96**. The openings may be formed using acceptable photolithography and etching techniques. After forming the openings for the source/drain contacts **112**, silicide layers **114** are formed through the openings for the source/drain contacts **112**. In some embodiments, a metallic material is deposited in the openings for the source/drain contacts **112**. The metallic material may comprise Ti, Co, Ni, NiCo, Pt, NiPt, Ir, PtIr, Er, Yb, Pd, Rh, Nb, a combination thereof, or the like, and may be formed using PVD, sputtering, a combination thereof, or the like. Subsequently, an annealing process is performed to form the silicide layers **114**. In some embodiments where the epitaxial source/drain regions **82** comprise silicon, the annealing process causes the metallic material to react with silicon to form a silicide of the metallic material at interfaces between the metallic material and the epitaxial source/drain regions **82**. After forming the silicide layers **114**, unreacted portions of the metallic material are removed using a suitable removal process. Subsequently, a liner, such as a diffusion barrier layer, an adhesion layer, or the like, and a conductive material are formed in the openings for the source/drain contacts **112** and in the openings for the gate contacts **110**. The liner may include titanium, titanium nitride, tantalum, tantalum nitride, a combination thereof, or the like. The conductive material may be copper, a copper alloy, silver, gold, tungsten, cobalt, aluminum, nickel, a combination thereof, or the like. A planarization process, such as a CMP process, may be performed to remove excess material from a surface of the second ILD **108**. The remaining portions of the liner and the conductive material form the source/drain contacts **112** and the gate contacts **110** in the openings. The source/drain contacts **112** are physically and electrically coupled to the respective epitaxial source/drain regions **82**, and the gate contacts **110** are physically and electrically coupled to the respective gate electrodes **94**. The source/drain contacts **112** and gate contacts **110** may be formed in different processes, or may be formed in the same process. Although shown as being formed in the same cross-sections, it should be appreciated that each of the source/drain contacts **112** and the gate contacts **110** may be formed in different cross-sections, which may avoid shorting of the contacts.

(49) FIGS. **20P**, **20N**, **21P**, **21N**, **22P**, **22N**, **23P**, **23N**, **24P**, **24N**, **25P**, **25N**, **26P**, **26N**, **27P**, and **27N** are cross-sectional views of intermediate stages in the manufacturing of gate stacks in accordance with some embodiments. FIGS. **20P-27P** illustrate detailed views of a region **89** of FIG. **14B** in the region **50P** (see FIGS. **2-7**). FIGS. **20N-27N** illustrate detailed views of a region **89** of FIG. **14B** in the region **50N** (see FIGS. **2-7**). In FIGS. **20P** and **20N**, an opening **90P** of the openings **90** (see FIG. **13B**) in the region **50P** and an opening **90N** of the openings **90** in the region **50N** are illustrated, respectively. As described below in greater detail, a gate stack **92P/94P** (including a gate dielectric layer **92P** and a corresponding overlying gate electrode **94P** as illustrated in FIG. **27P**) is formed in the opening **90P** and a gate stack **92N/94N** (including a gate dielectric layer **92N** and a corresponding overlying gate electrode **94N** as illustrated in FIG. **27N**) is formed in the opening **90N**.

(50) Further in FIGS. **20P** and **20N**, a gate dielectric layer **92P** is formed in the opening **90P** in the region **50P** and a gate dielectric layer **92N** is formed in the opening **90N** in the region **50N**, respectively. The gate dielectric layers **92P** and **92N** are deposited conformally in the openings **90P** and **90N**, respectively, such as on the top surfaces and the sidewalls of the fins **52** and on sidewalls of the gate seal spacers **80**/gate spacers **86**. The gate dielectric layers **92P** and **92N** may also be

formed on top surface of the first ILD **88**. In some embodiments, each of the gate dielectric layers **92P** and **92N** may comprise silicon oxide, silicon nitride, or multilayers thereof. In some embodiments, each of the gate dielectric layers **92N** and **92P** may comprise a high-k dielectric material, and in these embodiments, the gate dielectric layers **92P** and **92N** may have a k value greater than about 7.0, and may include a metal oxide or a silicate of Hf, Al, Zr, La, Mg, Ba, Ti, Pb, combinations thereof, or the like. The formation methods of the gate dielectric layers **92P** and **92N** may include Molecular-Beam Deposition (MBD), ALD, PECVD, a combination thereof, or the like. In embodiments where portions of the dummy dielectric layer **60** remain in the openings **90P** and **90N**, the gate dielectric layers **92P** and **92N** include a material of the dummy dielectric layer **60** (e.g., SiO.sub.2).

(51) In some embodiments, the formation of the gate dielectric layers **92P** in the region **50P** and the gate dielectric layers **92N** in the region **50N** may occur by simultaneously conformally depositing a same dielectric material in the openings **90P** and **90N**, respectively. In such embodiments, the gate dielectric layers **92P** and **92N** comprise the same dielectric material. In other embodiments, a process for forming the gate dielectric layers **92P** in the region **50P** may be different from and not simultaneous with a process for forming the gate dielectric layers **92N** in the region **50N**. In such embodiments, the gate dielectric layers **92P** and **92N** may comprise different dielectric materials. Various masking steps may be used to mask and expose appropriate ones of regions **50P** and **50N** when using non-simultaneous processes. In some embodiments, the gate dielectric layers **92P** have a thickness between about 8 Å and about 100 Å. In some embodiments, the gate dielectric layers **92N** have a thickness between about 8 Å and about 100 Å. In some embodiments when thicknesses of the gate dielectric layers **92P** and the gate dielectric layers **92N** are less than about 8 Å, such small thicknesses lead to high leakage currents through the gate dielectric layers **92P** and **92N**. In some embodiments when thicknesses of the gate dielectric layers **92P** and the gate dielectric layers **92N** are greater than about 100 Å, such large thicknesses lead to poor device speeds.

(52) In FIGS. **21P** and **21N**, a P-type work function layer **94.sub.1** is formed in the opening **90P** in the region **50P**. In some embodiments, a P-type work function material is formed in the opening **90P** and **90N** in both the region **50P** and the region **50N**. In such embodiments, the P-type work function material is subsequently removed from the region **50N**, such that the P-type work function material remains only in the region **50P** and forms P-type work function layer **94.sub.1**. In other embodiments, a mask layer is formed over the region **50N** and a P-type work function material is formed in the opening **90P** in the region **50P**. In such embodiments, the mask layer is removed after forming the P-type work function material in the opening **90P**. In some embodiments, the P-type work function layer **94.sub.1** may comprise TiN, WN, TaN, Ru, Co, combinations thereof, multi-layers thereof, or the like, and may be formed using PVD, CVD, ALD, a combination thereof, or the like. In some embodiments, the P-type work function layer **94.sub.1** has a thickness between about 5 Å and about 300 Å. In some embodiments when the thickness of the P-type work function layer **94.sub.1** is less than about 5 Å, such a small thickness leads to an insufficient work function value for the resulting device and to work function value and work function thickness non-uniformity across the wafer. In some embodiments when the thickness of the P-type work function layer **94.sub.1** is greater than about 300 Å, such a large thickness leads to high gate resistance (R.sub.G).

(53) In FIGS. **22P** and **22N**, an N-type work function layer **94.sub.2** is formed in the opening **90P** in the region **50P** and the opening **90N** in the region **50N**. In some embodiments, the N-type work function layer **94.sub.2** may comprise Ti, Ag, Al, TiAl, TiAlN, TiAlC, TaC, TaCN, TaSiN, TaAlC, Mn, Zr, combinations thereof, multi-layers thereof, or the like, and may be formed using PVD, CVD, ALD, a combination thereof, or the like. In some embodiments, the N-type work function layer **94.sub.2** has a thickness between about 8 Å and about 400 Å. In some embodiments when the thickness of the N-type work function layer **94.sub.2** is less than about 8 Å, such a small thickness leads to an insufficient work function value for the resulting device and to work function value and

work function thickness non-uniformity across the wafer. In some embodiments when the thickness of the N-type work function layer **94.sub.2** is greater than about 400 Å, such a large thickness leads to high gate resistance (R.sub.G).

(54) In FIGS. **23P** and **23N**, a protective layer **94.sub.3** is formed in the opening **90P** in the region **50P** and the opening **90N** in the region **50N**. In some embodiments, the protective layer **94.sub.3** may comprise Ti, Si, TiN, TiSi, TiAl, SiC, TaN, combinations thereof, multi-layers thereof, or the like, and may be formed using PVD, CVD, ALD, a combination thereof, or the like. In some embodiments, the protective layer **94.sub.3** protects the N-type work function layer **94.sub.2** from oxidizing during subsequent process steps. In some embodiments, the protective layer **94.sub.3** has a thickness between about 5 Å and about 100 Å. In some embodiments when the thickness of the protective layer **94.sub.3** is less than about 5 Å, such a small thickness is not able to provide sufficient protection to the P-type work function layer **94.sub.1** and to the N-type work function layer **94.sub.2** and also has poor thickness uniformity across the wafer. In some embodiments when the thickness of the protective layer **94.sub.3** is greater than about 100 Å, such a large thickness leads to high gate resistance (R.sub.G) and to effective work function degradation.

(55) In some embodiments where the openings **90P** and **90N** have widths of about 23 nm to about 300 nm, the layers, such as the gate dielectric layers **92P** and **92N**, the P-type work function layer **94.sub.1**, the N-type work function layer **94.sub.2**, and the protective layer **94.sub.3**, may not be deposited conformally in the openings **90P** and **90N**. In some embodiments, thicknesses of the layers disposed on upper portions of sidewall of the openings **90P** and **90N** may be greater than thicknesses of the layers disposed on lower portions of the sidewall of the openings **90P** and **90N**. Such a non-conformal deposition of the layers results in narrowing or pinching off of the upper portions of the openings **90P** and **90N**. The narrowing of the upper portions of the openings **90P** and **90N** may prevent additional layers that are subsequently deposited in the openings **90P** and **90N** from fully filling up the openings **90P** and **90N**. In some embodiments, due to the partial filling of the openings **90P** and **90N**, the resulting gate stacks may comprise seam void defects that adversely affect the functionality of the gate stacks. As described below in greater detail, an etching process is performed to widen the upper portions of the openings **90P** and **90N** that improves the filling properties of the openings **90P** and **90N** and reduce/avoid any seam void defects.

(56) In FIGS. **24P** and **24N**, an etch process (indicated by arrows **116**) is performed to widen the upper portions of the openings **90P** and **90N**. In some embodiments, the etch process **116** is a plasma process performed by a plasma generated from a gas mixture comprising Ar, He, H.sub.2, N.sub.2, CF.sub.4, Cl.sub.2, BCl.sub.3, a combination thereof, or the like. In some embodiments, the etch process **116** may be performed with a plasma power of between about 200 W and about 1400 W. In some embodiments, the etch process **116** may be performed with a substrate voltage bias of between about 30 V and about 250 V. In some embodiments, the etch process **116** may be performed at a pressure between about 4 mTorr and about 200 mTorr. In some embodiments, the etch process **116** may be performed at a temperature between about 20° C. and about 250° C. In some embodiments, the etch process **116** may be performed during an etch time of between about 5 s and about 350 s.

(57) In some embodiments, the etch process **116** removes portions of the protective layer **94.sub.3**, the N-type work function layer **94.sub.2**, and/or the P-type work function layer **94.sub.1** from sidewalls of the upper portions of the openings **90P** and **90N** to widen the upper portions of the openings **90P** and **90N**. In some embodiments, the amount of the removed material depends on parameters of the etch process **116** and may be tuned by tuning the plasma power, the substrate bias voltage, the pressure, the temperature, the etch time, and/or the plasma composition of the etch process **116**. In some embodiments, the etch process **116** completely removes the protective layer **94.sub.3** from the sidewalls of the upper portions of the openings **90P** and **90N**. In other embodiments, in addition to completely removing the protective layer **94.sub.3** from the sidewalls of the upper portions of the openings **90P** and **90N**, the etch process **116** removes portions of the P-

type work function layer **94.sub.1** form the sidewalls of the upper portion of the opening **90N** and portions of the P-type work function layer **94.sub.1** or portions of the N-type work function layer **94.sub.2** from the sidewalls of the upper portion of the opening **90P**.

(58) In some embodiments, after performing the etch process **116**, the N-type work function layer **94.sub.2** and the protective layer **94.sub.3** have non-uniform thicknesses along the sidewalls of the openings **90P** and **90N**. In some embodiments, the thickness of the protective layer **94.sub.3** decreases as the protective layer **94.sub.3** extends along the sidewalls of the openings **90P** and **90N** away from the respective channel regions **58**. In some embodiments, the thickness of the N-type work function layer **94.sub.2** decreases as the N-type work function layer **94.sub.2** extends along the sidewalls of the openings **90P** and **90N** away from the respective channel regions **58**. In some embodiments, the N-type work function layer **94.sub.2** may extend along the sidewalls of the openings **90P** and **90N** all the way to the top of the openings **90P** and **90N**. In other embodiments, the N-type work function layer **94.sub.2** may not reach the top of the openings **90P** and **90N** and may partially line the sidewalls of the openings **90P** and **90N**.

(59) In FIGS. **25P** and **25N**, after performing the etch process **116** (see FIGS. **24P** and **24N**), a protective layer **94.sub.4** is formed in the opening **90P** in the region **50P** and in the opening **90N** in the region **50N**. In some embodiments, the protective layer **94.sub.4** may be formed using similar materials and methods as the protective layer **94.sub.3** described above with reference to FIGS. **23P** and **23N**, and the description is not repeated herein. In some embodiments, the protective layer **94.sub.4** and the protective layer **94.sub.3** comprise a same material. In other embodiments, the protective layer **94.sub.4** and the protective layer **94.sub.3** comprise different materials. In some embodiments, the protective layer **94.sub.4** protects exposed portion of the N-type work function layer **94.sub.2** and/or exposed portions of the P-type work function layer **94.sub.1** from oxygen (O), carbon (C), and fluorine (F) pickup during subsequent processes steps. In some embodiments, the protective layer **94.sub.4** has a thickness between about 5 Å and about 100 Å. In some embodiments when the thickness of the protective layer **94.sub.4** is less than about 5 Å, such a small thickness is not able to provide sufficient protection to the P-type work function layer **94.sub.1** and the N-type work function layer **94.sub.2** and causes thickness non-uniformity across the wafer. In some embodiments when the thickness of the protective layer **94.sub.4** is greater than about 100 Å, such a large thickness leads to high gate resistance ($R_{sub.G}$) and to effective work function degradation.

(60) In FIGS. **26P** and **26N**, a glue layer **94.sub.5** is formed in the opening **90P** in the region **50P** and in the opening **90N** in the region **50N**. In some embodiments, the glue layer **94.sub.5** may comprise Ti, TiN, Co, W, combinations thereof, multilayers thereof, or the like and may be formed using PVD, CVD, ALD, a combination thereof, or the like. In some embodiments, the glue layer **94.sub.5** has a thickness between about 8 Å and about 90 Å. In some embodiments when the thickness of the glue layer **94.sub.5** is less than about 8 Å, the glue layer of such a small thickness may not fully cover the wafer and may adversely affect nucleation of a subsequently formed fill material **94.sub.6**. In some embodiments when the thickness of the glue layer **94.sub.5** is greater than about 90 Å, such a large thickness may lead to high gate resistance ($R_{sub.G}$). Subsequently, a fill material **94.sub.6** is formed in the opening **90P** in the region **50P** and in the opening **90N** in the region **50N**. In some embodiments, the fill material **94.sub.6** overfills the openings **90P** and **90N**, such that portions of the fill material **94.sub.6** extend along the top surface of the first ILD **88**. The fill material **94.sub.6** may comprise Co, Ru, Al, Ag, Au, W, Ni, Ti, Cu, Mn, Pd, Re, Ir, Pt, Zr, alloys thereof, combinations thereof, multi-layers thereof, or the like, and may be formed using PVD, CVD, ALD, plating, a combination thereof, or the like. By widening the openings **90P** and **90N** as described above with reference to FIGS. **21P** and **21N**, the fill material **94.sub.6** fully fills up the openings **90P** and **90N** without forming seam void defects within the fill material **94.sub.6**.

(61) By forming the fill material **94.sub.6** without seam void defects, various issues caused by the seam void defects may be avoided. For example, various issues caused by the seam void defects

include high gate resistance (R.sub.G) due to air gaps within the fill material **94.sub.6**, device speed degradation due to poor contact between the fill material **94.sub.6** and other layers of the gate stack, increase in gate resistance (R.sub.G) and degradation of work function and threshold voltage (V.sub.t) due to oxidation of gate stack layers caused by penetration of chemicals/gases into the seam void defects during a subsequently performed gate cut process.

(62) In FIGS. **27P** and **27N**, after forming the fill material **94.sub.6**, a planarization process, such as a CMP process, may be performed to remove excess portions of the layers **92P**, **92N**, **94.sub.1-94.sub.6**, which excess portions are over the top surface of the first ILD **88**. The remaining portion of the gate dielectric layer **92P** within the opening **90P** forms the gate dielectric **92P** of a replacement gates stack of the resulting FinFET device in the region **50P**. The remaining portions of the layers **94.sub.1-94.sub.6** within the opening **90P** form the gate electrode **94P** of the replacement gates stack of the resulting FinFET device in the region **50P**. The gate electrode **94P** and the respective gate dielectric **92P** form the gate stack **92P/94P** in the region **50P**. The remaining portion of the gate dielectric layer **92N** within the opening **90N** forms the gate dielectric **92N** of a replacement gates stack of the resulting FinFET device in the region **50N**. The remaining portions of the layers **94.sub.2-94.sub.6** within the opening **90N** form the gate electrode **94N** of the replacement gates stack of the resulting FinFET device in the region **50N**. The gate electrode **94N** and the respective gate dielectric **92N** form the gate stack **92N/94N** in the region **50N**. In some embodiments, the remaining portion of the fill material **94.sub.6** has a non-uniform width, such that the width of the remaining portion of the fill material **94.sub.6** decreases as the remaining portion of the fill material **94.sub.6** extends from the topmost surface of the ILD **88** (see FIG. **14B**) toward the respective channel region **58**.

(63) In some embodiments, after forming the gate stacks **92P/94P** and **92N/94N**, the gate stacks **92P/94P** and **92N/94N** may be patterned in a gate cut process. In some embodiments, the gate cut process may be performed as described above with reference to FIGS. **15A**, **15B**, **15C**, **15D**, **16A**, **16B**, **16C**, **16D**, **17A**, **17B**, **17C**, and **17D**, and the description is not repeated herein. By forming the gate stacks **92P/94P** and **92N/94N** as described above with reference to FIGS. **20P**, **20N**, **21P**, **21N**, **22P**, **22N**, **23P**, **23N**, **24P**, **24N**, **25P**, **25N**, **26P**, **26N**, **27P**, and **27N**, seam void defects are not formed in the gate stacks **92P/94P** and **92N/94N** and etchants of the gate cut process do not penetrate into the seam void defects to damage the gate stacks **92P/94P** and **92N/94N**.

(64) FIGS. **28P**, **28N**, **29P**, **29N**, **30P**, **30N**, **31P**, **31N**, **32P**, **32N**, **33P**, **33N**, **34P**, **34N**, **35P**, and **35N** are cross-sectional views of intermediate stages in the manufacturing of gate stacks in accordance with some embodiments. FIGS. **28P-35P** illustrate detailed views of a region **89** of FIG. **14B** in the region **50P** (see FIGS. **2-7**). FIGS. **28N-35N** illustrate detailed views of a region **89** of FIG. **14B** in the region **50N** (see FIGS. **2-7**). As described below in greater detail, a gate stack **92P/94P** (including a gate dielectric **92P** and a corresponding overlying gate electrode **94P** illustrated in FIG. **35P**) is formed in the opening **90P** in the region **50P** and a gate stack **92N/94N** (including a gate dielectric **92N** and a corresponding overlying gate electrode **94N** illustrated in FIG. **35N**) is formed in the opening **90N** in the region **50N**. In some embodiments, some process steps described below with reference to FIGS. **28P**, **28N**, **29P**, **29N**, **30P**, **30N**, **31P**, **31N**, **32P**, **32N**, **33P**, **33N**, **34P**, **34N**, **35P**, and **35N** are similar to some of the process steps described above with reference to FIGS. **20P**, **20N**, **21P**, **21N**, **22P**, **22N**, **23P**, **23N**, **24P**, **24N**, **25P**, **25N**, **26P**, **26N**, **27P**, and **27N**, with like features being labeled by like numerical references, and the description of the like features is not repeated herein.

(65) In FIGS. **28P** and **28N**, after forming a P-type work function layer **94.sub.1** in the opening **90P** in the region **50P** as described above with reference to FIGS. **21P** and **21N**, a protective layer **94.sub.3** is formed in the opening **90P** in the region **50P** and the opening **90N** in the region **50N** as described above with reference to FIGS. **23P** and **23N**, and the description is not repeated herein.

(66) In FIGS. **29P** and **29N**, an etch process (indicated by arrows **118**) is performed to widen upper portions of the openings **90P** and **90N**. In some embodiments, the etch process **118** is similar to the

etch process **116** described above with reference to FIGS. **24P** and **24N**, and the description is not repeated herein. In some embodiments, the etch process **118** removes portions of the gate dielectric layer **92N**, the P-type work function layer **94.sub.1**, or the protective layer **94.sub.3** from sidewalls of the upper portions of the openings **90P** and **90N** to widen the upper portions of the openings **90P** and **90N**. In some embodiments, the amount of the removed material depends on parameters of the etch process **118** and may be tuned by tuning the plasma power, the substrate bias voltage, the pressure, the temperature, the etch time, and/or the plasma composition of the etch process **118**. In some embodiments, the etch process **118** completely removes the protective layer **94.sub.3** from sidewalls of the upper portions of the openings **90P** and **90N**. In other embodiments, in addition to completely removing the protective layer **94.sub.3** from the sidewalls of the upper portions of the openings **90P** and **90N**, the etch process **118** further removes portions of the P-type work function layer **94.sub.1** from the sidewalls of the upper portions of the openings **90P** and portions of the gate dielectric layer **92N** from the sidewalls of the upper portions of the openings **90N**.

(67) In some embodiments, after performing the etch process **118**, the P-type work function layer **94.sub.1** and the protective layer **94.sub.3** have non-uniform thicknesses along the sidewalls of the openings **90P** and **90N**. In some embodiments, the thickness of the protective layer **94.sub.3** decreases as the protective layer **94.sub.3** extends along the sidewalls of the openings **90P** and **90N** away from the respective channel regions **58**. In some embodiments, the thickness of the P-type work function layer **94.sub.1** decreases as the P-type work function layer **94.sub.1** extends along the sidewalls of the opening **90P** away from the respective channel region **58**. In some embodiments, the P-type work function layer **94.sub.1** may extend along the sidewalls of the opening **90P** all the way to the top of the opening **90P**. In other embodiments, the P-type work function layer **94.sub.1** may not reach the top of the opening **90P** and may partially line the sidewalls of the opening **90P**.

(68) In FIGS. **30P** and **30N**, an N-type work function layer **94.sub.2** is formed in the opening **90P** in the region **50P** and the opening **90N** in the region **50N** as described above with reference to FIGS. **22P** and **22N**, and the description is not repeated herein.

(69) In FIGS. **31P** and **31N**, a protective layer **94.sub.4** is formed in the opening **90P** in the region **50P** and the opening **90N** in the region **50N** as described above with reference to FIGS. **25P** and **25N**, and the description is not repeated herein.

(70) In FIGS. **32P** and **32N**, another etch process (indicated by arrows **120**) is performed to widen the upper portions of the openings **90P** and **90N**. In some embodiments, the etch process **120** is similar to the etch process **116** described above with reference to FIGS. **24P** and **24N**, and the description is not repeated herein. In some embodiments, the etch process **120** removes portions of the N-type work function layer **94.sub.2** or portions of the protective layer **94.sub.4** from sidewalls of the upper portions of the openings **90P** and **90N** to widen the upper portions of the openings **90P** and **90N**. In some embodiments, the amount of the removed material depends on parameters of the etch process **120** and may be tuned by tuning the plasma power, the substrate bias voltage, the pressure, the temperature, the etch time, and/or the plasma composition of the etch process **120**. In some embodiments, the etch process **120** completely removes the protective layer **94.sub.4** from sidewalls of the upper portions of the openings **90P** and **90N**. In other embodiments, in addition to completely removing the protective layer **94.sub.4** from sidewalls of the upper portions of the openings **90P** and **90N**, the etch process **120** further removes portions of the N-type work function layer **94.sub.2** from the sidewalls of the upper portions of the openings **90P** and **90N**.

(71) In some embodiments, after performing the etch process **120**, the N-type work function layer **94.sub.2** and the protective layer **94.sub.4** have non-uniform thicknesses along the sidewalls of the openings **90P** and **90N**. In some embodiments, the thickness of the protective layer **94.sub.4** decreases as the protective layer **94.sub.4** extends along the sidewalls of the openings **90P** and **90N** away from the respective channel regions **58**. In some embodiments, the thickness of the N-type work function layer **94.sub.2** decreases as the N-type work function layer **94.sub.2** extends along

the sidewalls of the openings **90P** and **90N** away from the respective channel regions **58**. In some embodiments, the N-type work function layer **94.sub.2** may extend along the sidewalls of the openings **90P** and **90N** all the way to the top of the openings **90P** and **90N**. In other embodiments, the N-type work function layer **94.sub.2** may not reach the top of the openings **90P** and **90N** and may partially line the sidewalls of the openings **90P** and **90N**.

(72) In FIGS. **33P** and **33N**, a protective layer **94.sub.7** is formed in the opening **90P** in the region **50P** and the opening **90N** in the region **50N**. In some embodiments, the protective layer **94.sub.7** may be formed using similar materials and methods as the protective layer **94.sub.3** described above with reference to FIGS. **23P** and **23N**, and the description is not repeated herein.

(73) In FIGS. **34P** and **34N**, a glue layer **94.sub.5** is formed in the opening **90P** in the region **50P** and in the opening **90N** in the region **50N** as described above with reference to FIGS. **26P** and **26N**, and the description is not repeated herein. Subsequently, a fill material **94.sub.6** is formed in the opening **90P** in the region **50P** and in the opening **90N** in the region **50N** as described above with reference to FIGS. **26P** and **26N**, and the description is not repeated herein. By widening the openings **90P** and **90N** as described above with reference to FIGS. **29P**, **29N**, **32P**, and **32N**, the fill material **94.sub.6** fully fills up the openings **90P** and **90N** without forming seam void defects within the fill material **94.sub.6** in the openings **90P** and **90N**.

(74) In FIGS. **35P** and **35N**, after forming the fill material **94.sub.6**, a planarization process, such as a CMP process, may be performed to remove excess portions of the layers **92P**, **92N**, **94.sub.1-94.sub.7**, which excess portions are over the top surface of the first ILD **88**. The remaining portion of the gate dielectric layer **92P** within the opening **90P** forms a gate dielectric **92P** of a replacement gates stack of the resulting FinFET device in the region **50P**. The remaining portions of the layers **94.sub.1-94.sub.7** within the opening **90P** form the gate electrode **94P** of the replacement gates stack of the resulting FinFET device in the region **50P**. The gate electrode **94P** and the respective gate dielectric **92P** form the gate stack **92P/94P** in the region **50P**. The remaining portion of the gate dielectric layer **92N** within the opening **90N** forms a gate dielectric **92N** of a replacement gates stack of the resulting FinFET device in the region **50N**. The remaining portions of the layers **94.sub.2-94.sub.7** within the opening **90N** form the gate electrode **94N** of the replacement gates stack of the resulting FinFET device in the region **50N**. The gate electrode **94N** and the respective gate dielectric **92N** form the gate stack **92N/94N** in the region **50N**. In some embodiments, the remaining portion of the fill material **94.sub.6** has a non-uniform width, such that the width of the remaining portion of the fill material **94.sub.6** decreases as the remaining portion of the fill material **94.sub.6** extends from the topmost surface of the ILD **88** (see FIG. **14B**) toward the respective channel region **58**.

(75) In some embodiments, after forming the gate stacks **92P/94P** and **92N/94N**, the gate stacks **92P/94P** and **92N/94N** may be patterned in a gate cut process. In some embodiments, the gate cut process may be performed as described above with reference to FIGS. **15A**, **15B**, **15C**, **15D**, **16A**, **16B**, **16C**, **16D**, **17A**, **17B**, **17C**, and **17D**, and the description is not repeated herein. By forming the gate stacks **92P/94P** and **92N/94N** as described above with reference to FIGS. **28P**, **28N**, **29P**, **29N**, **30P**, **30N**, **31P**, **31N**, **32P**, **32N**, **33P**, **33N**, **34P**, **34N**, **35P**, and **35N**, seam void defects are not formed in the gate stacks **92P/94P** and **92N/94N** and etchants of the gate cut process do not penetrate into the seam void defects to damage the gate stacks **92P/94P** and **92N/94N**.

(76) FIGS. **36P**, **36N**, **37P**, **37N**, **38P**, **38N**, **39P**, **39N**, **40P**, **40N**, **41P**, and **41N** are cross-sectional views of intermediate stages in the manufacturing of gate stacks in accordance with some embodiments. FIGS. **36P-41P** illustrate detailed views of a region **89** of FIG. **14B** in the region **50P** (see FIGS. **2-7**). FIGS. **36N-41N** illustrate detailed views of a region **89** of FIG. **14B** in the region **50N** (see FIGS. **2-7**). As described below in greater detail, a gate stack **92P/94P** (including a gate dielectric layer **92P** and a corresponding overlying gate electrode **94P** illustrated in FIG. **41P**) is formed in the opening **90P** in the region **50P** and a gate stack **92N/94N** (including a gate dielectric layer **92N** and a corresponding overlying gate electrode **94N** illustrated in FIG. **41N**) is formed in

the opening **90N** in the region **50N**. In some embodiments, some process steps described below with reference to FIGS. **36P**, **36N**, **37P**, **37N**, **38P**, **38N**, **39P**, **39N**, **40P**, **40N**, **41P**, and **41N** are similar to some of the process steps described above with reference to FIGS. **20P**, **20N**, **21P**, **21N**, **22P**, **22N**, **23P**, **23N**, **24P**, **24N**, **25P**, **25N**, **26P**, **26N**, **27P**, and **27N**, with like features being labeled by like numerical references, and the description of the like features is not repeated herein.

(77) In FIGS. **36P** and **36N**, after forming an N-type work function layer **94.sub.2** in the opening **90P** in the region **50P** and the opening **90N** in the region **50N** as described above with reference to FIGS. **22P** and **22N**, a sacrificial layer **122** is formed in the opening **90P** in the region **50P** and the opening **90N** in the region **50N**. In some embodiments, the sacrificial layer **122** overfills the openings **90P** and **90N** such that a portion of the sacrificial layer **122** extends along the top surface of the first ILD **88**. In some embodiments, the sacrificial layer **122** may comprise an organic photo sensitive cross linking polymer material, such as an organic backside anti-reflective coating (BARC), photoresist, or the like, and may be formed using a spin coating process, or the like. In some embodiments, adhesion promoters (such as, for example, Hexamethyl disilazane (HMDS)) and a hard mask layer (for example, comprising TiN) are formed before forming the sacrificial layer **122**. In some embodiments when the sacrificial layer **122** comprises organic BARC, the sacrificial layer **122** is formed by spin coating process followed by a baking process to remove solvent. In some embodiments, the sacrificial layer **122** has a thickness between about 300 nm to about 2000 nm.

(78) In FIGS. **37P** and **37N**, an etch process (indicated by arrows **124**) is performed to etch back the sacrificial layer **122** below the top surface of the first ILD **88**. In some embodiments, the etch process **124** is a plasma process performed by a plasma generated from a gas mixture comprising H.sub.2, N.sub.2, a combination thereof, or the like. In some embodiments, the etch process **124** may be performed with a plasma power of between about 20 W and about 300 W. In some embodiments, the etch process **124** may be performed with a substrate voltage bias of between about 25 V and about 150 V. In some embodiments, the etch process **124** may be performed at a pressure between about 8 mTorr and about 100 mTorr. In some embodiments, the etch process **124** may be performed at a temperature between about 20° C. and about 200° C. In some embodiments, the etch process **124** may be performed during an etch time between about 10 s and about 400 s.

(79) In some embodiments, the parameters of the etch process **124** may be tuned such that the sacrificial layer **122** is selectively etched without significantly etching the N-type work function layer **94.sub.2**. In some embodiments, in the region **50P**, a top surface of the sacrificial layer **122** is recessed below the top surface of the first ILD **88** by a distance D.sub.P. In some embodiments, the distance D.sub.P is between about 32 nm and about 80 nm. In some embodiments, the distance D.sub.P is between about 20% and about 75% of a height H.sub.P of the opening **90P**. In some embodiments, in the region **50N**, a top surface of the sacrificial layer **122** is recessed below the top surface of the first ILD **88** by a distance D.sub.N. In some embodiments, the distance D.sub.N is between about 32 nm and about 80 nm. In some embodiments, the distance D.sub.N is between about 20% and about 75% of a height H.sub.N of the opening **90N**.

(80) In FIGS. **38P** and **38N**, an etch process (indicated by arrows **126**) is performed to selectively remove portions of the P-type work function layer **94.sub.1** and the N-type work function layer **94.sub.2**, widen the upper portions of the openings **90P** and **90N** and improve the filling properties of the openings **90P** and **90N**. In some embodiments, the etch process **126** is stopped at the gate dielectric layer **92P**, when portions of the P-type work function layer **94.sub.1** and the N-type work function layer **94.sub.2** above a top surface of the recessed sacrificial layer **122** are completely removed. In some embodiments, after completing the etch process **126**, a top surface of the P-type work function layer **94.sub.1** and a top surface of the N-type work function layer **94.sub.2** are substantially level with the top surface of the recessed sacrificial layer **122**. In some embodiments, the etch process **126** is a wet etch process, which may be performed using a mixture comprising HCl, H.sub.2O, HF, NH.sub.4OH, H.sub.3PO.sub.4, (de-ionized) DI water, or the like. In the

embodiment illustrated in FIGS. 38P and 38N, the etch process 126 may also remove a portion of the gate dielectric layer 92P, such that slopes of exposed surfaces of the gate dielectric layer 92P in the openings 90P and 90N are altered compared to respective surfaces shown in FIGS. 37P and 37N. In other embodiments, the etch process 126 has a high selectivity for removing the P-type work function layer 94.sub.1 and the N-type work function layer 94.sub.2, without substantially etching of gate dielectric layer 92P. In such embodiments, the slopes of the exposed surfaces of the gate dielectric layer 92P in the openings 90P and 90N are not substantially altered and are substantially same as shown in FIGS. 37P and 37N.

(81) In some embodiments when distance D.sub.N (D.sub.P) is less than about 20% of the height H.sub.N (H.sub.P) of the opening 90N (90P), such a small distance does not allow for sufficient widening of the top portion of the opening 90N (90P) to avoid issues associated with seam void defects. In some embodiments when distance D.sub.N (D.sub.P) is greater than about 75% of the height H.sub.N (H.sub.P) of the opening 90N (90P), such a large distance causes degradation of effective work function, increase in threshold voltage (V.sub.t), decrease in saturation current (I.sub.sat), and poor device turn-on performance due to loss of gate stack layers (such as removed portions of the P-type work function layer 94.sub.1 and the N-type work function layer 94.sub.2).

(82) In FIGS. 39P and 39N, the recessed sacrificial layer 122 (see FIGS. 38P and 38N) is removed. In some embodiments, the recessed sacrificial layer 122 may be removed using an ashing process followed by a wet clean process. In some embodiments, the ashing process is a plasma process performed by plasma generated from a gas mixture comprising N.sub.2, H.sub.2, a combination thereof, or the like. In some embodiments when the ashing process is a plasma process, the plasma process is performed without any substrate bias. In some embodiments when the ashing process is a plasma process, the plasma process is performed at a temperature between about 50° C. and about 300° C. In some embodiments, the wet clean process may be performed using a mixture comprising DI water, dilute hydrofluoric (dHF) acid, a combination thereof, or the like.

(83) In FIGS. 40P and 40N, a glue layer 94.sub.5 is formed in the opening 90P in the region 50P and in the opening 90N in the region 50N as described above with reference to FIGS. 26P and 26N, and the description is not repeated herein. Subsequently, a fill material 94.sub.6 is formed in the opening 90P in the region 50P and in the opening 90N in the region 50N as described above with reference to FIGS. 26P and 26N, and the description is not repeated herein. By widening the openings 90P and 90N as described above with reference to FIGS. 38P and 38N, the fill material 94.sub.6 fully fills the openings 90P and 90N without forming seam void defects within the fill material 94.sub.6 in the openings 90P and 90N.

(84) In FIGS. 41P and 41N, after forming the fill material 94.sub.6, a planarization process, such as a CMP process, may be performed to remove excess portions of the layers 92P, 92N, 94.sub.5, and 94.sub.6, which excess portions are over the top surface of the first ILD 88. The remaining portion of the gate dielectric layer 92P within the opening 90P forms the gate dielectric 92P of a replacement gates stack of the resulting FinFET device in the region 50P. The remaining portions of the layers 94.sub.1, 94.sub.2, 94.sub.5, and 94.sub.6 within the opening 90P form the gate electrode 94P of the replacement gates stack of the resulting FinFET device in the region 50P. The gate electrode 94P and the respective gate dielectric 92P form the gate stack 92P/94P in the region 50P. The remaining portion of the gate dielectric layer 92N within the opening 90N form the gate dielectrics 92N of a replacement gates stack of the resulting FinFET device in the region 50N. The remaining portions of the layers 94.sub.2, 94.sub.5, and 94.sub.6 within the opening 90N form the gate electrode 94N of the replacement gates stack of the resulting FinFET device in the region 50N. The gate electrode 94N and the respective gate dielectric 92N form the gate stack 92N/94N in the region 50N.

(85) In some embodiments, after forming the gate stacks 92P/94P and 92N/94N, the gate stacks 92P/94P and 92N/94N may be patterned in a gate cut process. In some embodiments, the gate cut process may be performed as described above with reference to FIGS. 15A, 15B, 15C, 15D, 16A,

16B, 16C, 16D, 17A, 17B, 17C, and 17D, and the description is not repeated herein. By forming the gate stacks **92P/94P** and **92N/94N** as described above with reference to FIGS. **36P**, **36N**, **37P**, **37N**, **38P**, **38N**, **39P**, **39N**, **40P**, **40N**, **41P**, and **41N**, seam void defects are not formed in the gate stacks **92P/94P** and **92N/94N** and etchants of the gate cut process do not penetrate into the seam void defects to damage the gate stacks **92P/94P** and **92N/94N**.

(86) FIG. **42** is a flow diagram illustrating a method **4200** of forming a semiconductor device in accordance with some embodiments. The method **4200** may be used to form a gate stack in both a p-type device region and an n-type device region of a semiconductor device. In some embodiments when the method **4200** forms a gate stack in an n-type device region of a semiconductor device, all process steps of the method **4200** are performed. In some embodiments when the method **4200** forms a gate stack in a p-type device region of a semiconductor device, all process steps of the method **4200** are performed except for the step **4209**. The method **4200** starts with step **4201**, where a sacrificial gate (such as the gate **72** illustrated in FIGS. **8A** and **8B**) is formed over an active region of a substrate (such as the substrate **50** illustrated in FIGS. **8A** and **8B**) as described above with reference to **8A** and **8B**. In step **4203**, the sacrificial gate is removed to form an opening (such as the opening **90** illustrated in FIGS. **13A** and **13B**) as described above with reference to **13A** and **13B**. In step **4205**, a gate dielectric layer (such as the gate dielectric layer **92P** illustrated in FIG. **20P**) is formed in the opening as described above with reference to FIG. **20P**. In step **4207**, a P-type work function layer (such as the P-type work function layer **94.sub.1** illustrated in FIG. **21P**) is formed in the opening as described above with reference to FIG. **21P**. In some embodiments when the method **4200** forms a gate stack in a p-type device region of a semiconductor device, step **4209** is omitted. In some embodiments when the method **4200** forms a gate stack in an n-type device region of a semiconductor device, step **4209** is performed where the P-type work function layer is removed from the opening as described above with reference to FIG. **21N**. In step **4211**, an N-type work function layer (such as the N-type work function layer **94.sub.2** illustrated in FIG. **22P**) is formed in the opening as described above with reference to FIG. **22P**. In step **4213**, a first protective layer (such as the protective layer **94.sub.3** illustrated in FIG. **23P**) is formed in the opening as described above with reference to FIG. **23P**. In step **4215**, an etch process (such as the etch process **116** illustrated in FIG. **24P**) is performed to widen an upper portion of the opening as described above with reference to FIG. **24P**. In step **4217**, a second protective layer (such as the protective layer **94.sub.4** illustrated in FIG. **25P**) is formed in the opening as described above with reference to FIG. **25P**. In step **4219**, the opening is filled with a conductive material (such as the glue layer **94.sub.5** and the fill material **94.sub.6** illustrated in FIG. **26P**) as described above with reference to FIG. **26P**. In step **4221**, a gate cut process is performed as described above with reference to FIGS. **15A**, **15B**, **15C**, **15D**, **16A**, **16B**, **16C**, **16D**, **17A**, **17B**, **17C**, and **17D**.

(87) FIG. **43** is a flow diagram illustrating a method **4300** of forming a semiconductor device in accordance with some embodiments. The method **4300** may be used to form a gate stack in both a p-type device region and an n-type device region of a semiconductor device. In some embodiments when the method **4300** forms a gate stack in an n-type device region of a semiconductor device, all process steps of the method **4300** are performed. In some embodiments when the method **4300** forms a gate stack in a p-type device region of a semiconductor device, all process steps of the method **4300** are performed except for the step **4309**. The method **4300** starts with step **4301**, where a sacrificial gate (such as the gate **72** illustrated in FIGS. **8A** and **8B**) is formed over an active region of a substrate (such as the substrate **50** illustrated in FIGS. **8A** and **8B**) as described above with reference to **8A** and **8B**. In step **4303**, the sacrificial gate is removed to form an opening (such as the opening **90** illustrated in FIGS. **13A** and **13B**) as described above with reference to **13A** and **13B**. In step **4305**, a gate dielectric layer (such as the gate dielectric layer **92P** illustrated in FIG. **20P**) is formed in the opening as described above with reference to FIG. **20P**. In step **4307**, a P-type work function layer (such as the P-type work function layer **94.sub.1** illustrated in FIG. **21P**) is formed in the opening as described above with reference to FIG. **21P**. In some embodiments

when the method **4300** forms a gate stack in a p-type device region of a semiconductor device, step **4309** is omitted. In some embodiments when the method **4300** forms a gate stack in an n-type device region of a semiconductor device, step **4309** is performed where the P-type work function layer is removed from the opening as described above with reference to FIG. **21N**. In step **4311**, a first protective layer (such as the protective layer **94.sub.3** illustrated in FIG. **28P**) is formed in the opening as described above with reference to FIG. **28P**. In step **4313**, a first etch process (such as the etch process **118** illustrated in FIG. **29P**) is performed to widen an upper portion of the opening as described above with reference to FIG. **29P**. In step **4315**, an N-type work function layer (such as the N-type work function layer **94.sub.2** illustrated in FIG. **30P**) is formed in the opening as described above with reference to FIG. **30P**. In step **4317**, a second protective layer (such as the protective layer **94.sub.4** illustrated in FIG. **31P**) is formed in the opening as described above with reference to FIG. **31P**. In step **4319**, a second etch process (such as the etch process **120** illustrated in FIG. **32P**) is performed to widen an upper portion of the opening as described above with reference to FIG. **32P**. In step **4321**, a third protective layer (such as the protective layer **94.sub.7** illustrated in FIG. **33P**) is formed in the opening as described above with reference to FIG. **33P**. In step **4323**, the opening is filled with a conductive material (such as the glue layer **94.sub.5** and the fill material **94.sub.6** illustrated in FIG. **34P**) as described above with reference to FIG. **34P**. In step **4325**, a gate cut process is performed as described above with reference to FIGS. **15A**, **15B**, **15C**, **15D**, **16A**, **16B**, **16C**, **16D**, **17A**, **17B**, **17C**, and **17D**.

(88) FIG. **44** is a flow diagram illustrating a method **4400** of forming a semiconductor device in accordance with some embodiments. The method **4400** may be used to form a gate stack in both a p-type device region and an n-type device region of a semiconductor device. In some embodiments when the method **4400** forms a gate stack in an n-type device region of a semiconductor device, all process steps of the method **4400** are performed. In some embodiments when the method **4400** forms a gate stack in a p-type device region of a semiconductor device, all process steps of the method **4400** are performed except for the step **4409**. The method **4400** starts with step **4401**, where a sacrificial gate (such as the gate **72** illustrated in FIGS. **8A** and **8B**) is formed over an active region of a substrate (such as the substrate **50** illustrated in FIGS. **8A** and **8B**) as described above with reference to **8A** and **8B**. In step **4403**, the sacrificial gate is removed to form an opening (such as the opening **90** illustrated in FIGS. **13A** and **13B**) as described above with reference to **13A** and **13B**. In step **4405**, a gate dielectric layer (such as the gate dielectric layer **92P** illustrated in FIG. **20P**) is formed in the opening as described above with reference to FIG. **20P**. In step **4407**, a P-type work function layer (such as the N-type work function layer **94.sub.1** illustrated in FIG. **21P**) is formed in the opening as described above with reference to FIG. **21P**. In some embodiments when the method **4400** forms a gate stack in a p-type device region of a semiconductor device, step **4409** is omitted. In some embodiments when the method **4400** forms a gate stack in an n-type device region of a semiconductor device, step **4409** is performed where the P-type work function layer is removed from the opening as described above with reference to FIG. **21N**. In step **4411**, an N-type work function layer (such as the N-type work function layer **94.sub.2** illustrated in FIG. **22P**) is formed in the opening as described above with reference to FIG. **22P**. In step **4413**, a sacrificial layer (such as the sacrificial layer **122** illustrated in FIG. **36P**) is formed in the opening as described above with reference to FIG. **36P**. In step **4415**, the sacrificial layer is etched back as described above with reference to FIG. **37P**. In step **4417**, portions of the P-type work function layer and portions of the P-type work function layer disposed above remaining portions of the sacrificial layer are removed as described above with reference to FIG. **38P**. In step **4419**, the remaining portions of the sacrificial layer are removed as described above with reference to FIG. **39P**. In step **4421**, the opening is filled with a conductive material (such as the glue layer **94.sub.5** and the fill material **94.sub.6** illustrated in FIG. **40P**) as described above with reference to FIG. **40P**. In step **4423**, a gate cut process is performed as described above with reference to FIGS. **15A**, **15B**, **15C**, **15D**, **16A**, **16B**, **16C**, **16D**, **17A**, **17B**, **17C**, and **17D**.

(89) In accordance with an embodiment, a method includes forming a sacrificial gate over an active region of a substrate. The sacrificial gate is removed to form an opening. A gate dielectric layer is formed on sidewalls and a bottom of the opening. A first work function layer is formed over the gate dielectric layer in the opening. A first protective layer is formed over the first work function layer in the opening. A first etch process is performed to widen an upper portion of the opening. The opening is filled with a conductive material.

(90) In accordance with another embodiment, a method includes forming a sacrificial gate over an active region of a substrate. The sacrificial gate is removed to form an opening. A gate dielectric layer is formed on sidewalls and a bottom of the opening. A first work function layer is formed over the gate dielectric layer in the opening. The opening is filled with a sacrificial layer. The sacrificial layer is etched back to expose an upper portion of the first work function layer within the opening. The upper portion of the first work function layer is removed to expose an upper portion of the gate dielectric layer within the opening. A remaining portion of the sacrificial layer is removed. The opening is filled with a conductive material.

(91) In accordance with yet another embodiment, a device includes a gate stack over an active region of a substrate. The gate stack includes a conductive material, a first protective layer lining sidewalls and a bottom surface of the conductive material, and a second protective layer lining sidewalls and a bottom surface of the first protective layer. The first protective layer and the second protective layer include different materials. The gate stack further includes an N-type work function layer lining sidewalls and a bottom surface of the second protective layer. An upper portion of the N-type work function layer is in physical contact with an upper portion of the first protective layer. The gate stack further includes a gate dielectric layer lining sidewalls and a bottom surface of the N-type work function layer.

(92) In accordance with yet another embodiment, a device includes a gate stack over an active region of a substrate. The gate stack includes a conductive material. The conductive material comprising an upper portion having a first width and a lower portion having a second width. The first width being greater than the second width. An N-type work function layer and P-type work function layer line sidewalls and a bottom surface of the lower portion of the conductive material. The N-type work function layer is interposed between the P-type work function layer and the lower portion of the conductive material. Topmost surfaces of the P-type work function layer and the N-type work function layer are in physical contact with a bottom surface of the upper portion of the conductive material. A gate dielectric layer lines the sidewalls and the bottom surface of the lower portion of the conductive material and sidewalls of the upper portion of the conductive material. The gate dielectric layer is in physical contact with the P-type work function layer and the sidewalls of the upper portion of the conductive material.

(93) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A device, comprising: a gate stack over an active region of a substrate, the gate stack comprising: a gate dielectric layer over the active region; a P-type work function layer on the gate dielectric layer; an N-type work function layer over the P-type work function layer; a first protective layer over the N-type work function layer; a second protective layer over the first protective layer, an

- upper portion of the N-type work function layer being in physical contact with an upper portion of the second protective layer; and a conductive material over the second protective layer.
2. The device of claim 1, wherein an upper surface of the P-type work function layer is level with an upper surface of the gate dielectric layer.
 3. The device of claim 1, wherein an upper surface of the N-type work function layer is level with an upper surface of the gate dielectric layer.
 4. The device of claim 1, wherein the conductive material comprises an adhesion layer on the second protective layer and a fill material on the adhesion layer.
 5. The device of claim 1, wherein the first protective layer completely covers a bottom of the second protective layer in a cross-sectional view.
 6. The device of claim 1, wherein a thickness of the N-type work function layer decreases as the N-type work function layer extends toward an upper surface of the N-type work function layer.
 7. The device of claim 1, wherein a thickness of the first protective layer decreases as the first protective layer extends away from the substrate.
 8. A device, comprising: a gate stack over an active region of a substrate, the gate stack comprising: a gate dielectric layer over the active region; a P-type work function layer on the gate dielectric layer; a first protective layer over the P-type work function layer; an N-type work function layer over the first protective layer, an upper portion of the N-type work function layer being in physical contact with an upper portion of the P-type work function layer; and a conductive material over the N-type work function layer.
 9. The device of claim 8, further comprising a second protective layer over the N-type work function layer, a thickness of the second protective layer decreasing as the second protective layer extends away from the substrate.
 10. The device of claim 9, further comprising a third protective layer over the second protective layer.
 11. The device of claim 10, wherein the upper portion of the N-type work function layer is in physical contact with an upper portion of the second protective layer.
 12. The device of claim 8, wherein an upper surface of the gate dielectric layer is level with an upper surface of the P-type work function layer.
 13. The device of claim 12, wherein the upper surface of the gate dielectric layer is level with an upper surface of the N-type work function layer.
 14. The device of claim 8, wherein a thickness of the N-type work function layer decreases as the N-type work function layer extends away from the substrate.
 15. The device of claim 8, wherein a thickness of an upper portion of the gate dielectric layer is greater than a thickness of a lower portion of the gate dielectric layer.
 16. A device, comprising: a gate stack over an active region of a substrate, the gate stack comprising: a gate dielectric layer over the substrate, wherein a thickness of an upper portion of the gate dielectric layer is greater than a thickness of a lower portion of the gate dielectric layer; a P-type work function layer over the gate dielectric layer; an N-type work function layer over the P-type work function layer; and a conductive layer over the N-type work function layer, the conductive layer comprising an upper region and a lower region, a width of the upper region being greater than a width of the lower region, wherein an upper surface of the conductive layer is level with an upper surface of the gate dielectric layer, wherein the conductive layer directly contacts and completely covers the P-type work function layer and the N-type work function layer.
 17. The device of claim 16, wherein an upper portion of the gate dielectric layer is in physical contact with the conductive layer.
 18. The device of claim 16, wherein the conductive layer comprises a conductive lining layer and a fill material over the conductive lining layer.
 19. The device of claim 16, wherein a width of the conductive layer increases as the conductive layer extends toward the substrate from an upper surface of the P-type work function layer.

20. The device of claim 19, wherein a width of the conductive layer decreases as the conductive layer extends from the upper surface of the P-type work function layer toward an uppermost surface of the conductive layer.
